

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	U	V	W	X	Y	Z	AA	AI	
1	365 DataScience RE California Database																										
2	Practical example																										
3																											
4	Product										Customer																
5	ID	Building	Year of sale	Month of sale	Type of property	Property #	Area (ft.)	Price	Status	Customer ID	Entity	Name	Surname	Age at time of purchase	Interval	Y	Gender	Country	State	Purpose	Deal satisfaction	Mortgage	Source				
6	1030	1	2005	11	Apartment	30	743.09	\$246,172.68	Sold	C0028	Individual	Madalyn	Mercer	19	18-25	1986	F	USA	California	Home		5	No	Website			
7	1029	1	2005	10	Apartment	29	756.21	\$246,331.90	Sold	C0027	Individual	Lara	Carrillo	22	18-25	1983	F	USA	California	Home		5	No	Website			
8	2002	2	2007	7	Apartment	2	587.28	\$209,280.91	Sold	C0112	Individual	Donavan	Flowers	22	18-25	1985	M	USA	California	Home		1	Yes	Client			
9	2031	2	2007	12	Apartment	31	1604.75	\$452,667.01	Sold	C0160	Individual	Darien	Dorsey	22	18-25	1985	M	USA	California	Investment		3	Yes	Website			
10	1049	1	2004	11	Apartment	49	1375.45	\$467,083.31	Sold	C0014	Individual	Alessandra	Perry	25	18-25	1979	F	USA	California	Home		4	No	Agency			
11	3011	3	2007	9	Apartment	11	675.19	\$203,491.85	Sold	C0125	Individual	Kaitlin	Owen	26	26-35	1981	F	USA	Virginia	Investment		5	No	Client			
12	3026	3	2007	9	Apartment	26	670.89	\$212,520.83	Sold	C0125	Individual	Kaitlin	Owen	26	26-35	1981	F	USA	Virginia	Investment		5	No	Agency			
13	3023	3	2008	1	Apartment	23	720.81	\$198,591.85	Sold	C0166	Individual	Terry	Forbes	26	26-35	1982	M	USA	California	Home		5	No	Client			
14	1031	1	2006	6	Apartment	31	782.25	\$265,467.68	Sold	C0034	Individual	Kole	Shannon	27	26-35	1979	M	USA	Arizona	Home		2	Yes	Website			
15	4023	4	2006	3	Apartment	23	794.52	\$235,633.26	Sold	C0170	Individual	Emmy	Singh	27	26-35	1979	F	USA	Virginia	Investment		3	Yes	Agency			
16	1036	1	2004	10	Apartment	36	1160.36	\$317,473.86	Sold	C0009	Individual	Arabella	Ferrell	28	26-35	1976	F	USA	Oregon	Home		1	No	Agency			
17	1046	1	2006	8	Apartment	46	1942.50	\$503,790.23	Sold	C0041	Individual	Christian	Costa	26	26-35	1980	M	USA	California	Home		5	No	Website			
18	4035	4	2007	10	Apartment	35	794.52	\$217,786.38	Sold	C0057	Individual	Michelle	Cameron	29	26-35	1978	F	USA	Nevada	Home		3	Yes	Website			
19	2036	2	2006	11	Apartment	36	1109.25	\$460,001.26	Sold	C0061	Individual	Enrique	Cardenas	29	26-35	1977	M	USA	California	Home		2	No	Website			
20	2056	2	2007	4	Apartment	56	1400.95	\$460,001.26	Sold	C0089	Individual	Amanda	Simon	29	26-35	1978	F	USA	California	Home		5	No	Agency			
21	1047	1	2007	12	Apartment	47	1479.72	\$448,134.27	Sold	C0159	Individual	Kamden	Stewart	29	26-35	1978	M	USA	California	Home		5	No	Website			
22	5051	5	2006	3	Apartment	51	790.54	\$249,591.99	Sold	C0171	Individual	Skylar	Buchanan	29	26-35	1977	M	USA	Nevada	Home		4	Yes	Website			
23	2007	2	2006	8	Apartment	7	723.93	\$196,142.19	Sold	C0042	Individual	Michael	Juarez	30	26-35	1976	M	USA	Colorado	Home		4	Yes	Agency			
24	3020	3	2007	4	Apartment	20	781.07	\$258,572.48	Sold	C0093	Individual	Antonio	Porter	30	26-35	1977	M	USA	Arizona	Home		5	No	Client			
25	3029	3	2007	4	Apartment	29	1127.76	\$310,831.21	Sold	C0093	Individual	Antonio	Porter	30	26-35	1977	M	USA	Arizona	Home		3	No	Client			
26	3015	3	2006	10	Apartment	15	720.70	\$207,281.59	Sold	C0051	Individual	Conner	Huff	31	26-35	1975	M	USA	Nevada	Home		5	No	Website			
27	2004	2	2006	12	Apartment	4	649.69	\$168,834.04	Sold	C0064	Individual	Joaquin	Mullins	31	26-35	1975	M	USA	California	Investment		5	No	Agency			
28	2006	2	2006	12	Apartment	6	1307.45	\$396,973.83	Sold	C0064	Individual	Joaquin	Mullins	31	26-35	1975	M	USA	California	Investment		5	No	Agency			
29	5013	5	2007	9	Apartment	13	618.38	\$188,743.11	Sold	C0128	Individual	Kyla	Walker	31	26-35	1976	F	USA	Colorado	Home		4	Yes	Agency			
30	1026	1	2005	3	Apartment	26	625.80	\$179,674.08	Sold	C0019	Individual	Victor	Jensen	32	26-35	1973	M	USA	California	Home		4	No	Website			
31	2054	2	2006	6	Apartment	54	1203.29	\$306,363.64	Sold	C0037	Individual	Tyler	Carr	32	26-35	1974	M	USA	California	Home		5	Yes	Client			
32	3033	3	2007	9	Apartment	33	670.89	\$200,300.63	Sold	C0127	Individual	Maia	Chandler	32	26-35	1975	F	USA	Utah	Home		1	Yes	Website			
33	1025	1	2005	3	Apartment	25	1434.09	\$382,041.13	Sold	C0018	Individual	Rodrigo	Ramirez	33	26-35	1972	M	USA	California	Home		1	No	Agency			
34	3027	3	2006	8	Apartment	27	781.07	\$245,572.79	Sold	C0040	Individual	Gianni	Fritz	33	26-35	1973	M	USA	California	Home		3	No	Website			
35	3031	3	2007	3	Apartment	31	1596.35	\$407,214.29	Sold	C0080	Individual	Janelle	Espinoza	33	26-35	1974	F	USA	California	Investment		4	No	Client			
36	2043	2	2007	4	Apartment	43	1110.32	\$355,073.40	Sold	C0083	Individual	Julio	Leonard	33	26-35	1974	M	USA	Nevada	Home		2	Yes	Website			
37	3034	3	2007	4	Apartment	34	781.07	\$256,821.64	Sold	C0085	Individual	Melanie	Holland	33	26-35	1974	F	USA	California	Investment		3	No	Client			
38	3016	3	2007	4	Apartment	16	697.89	\$226,342.80	Sold	C0091	Individual	Gordon	Brown	33	26-35	1974	M	UK	Arizona	Investment		4	No	Agency			
39	1018	1	2004	10	Apartment	18	625.80	\$191,389.87	Sold	C0007	Individual	Jaelyn	Berger	34	26-35	1970	F	USA	California	Home		3	No	Website			

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1	365 DataScience RE California Database																				
2	Types of data and levels of measurement																				
3																					
4	Task 1: What are the types of data and the levels of measurement of the following variables: Cust ID, Mortgage, Year of sale																				
5																					
6																					
7		Type of data		Level of measurement																	
8	Cust ID	Numerical		Ratio																	
9	Mortgage	Categorical		Nominal																	
10	Year of sale	Categorical		Ordinal																	
11																					
12																					
13	Variable	Type of data	Level of measurement		Comment																
14	Cust ID	Categorical	Nominal		This variable has the same properties as ID.																
15	Mortgage	Categorical	Nominal		This is a Binary variable. Like a Yes/No question or Gender.																
16	Year of sale	Numerical, discrete		Interval		Year is a numerical variable. It is always discrete. The level of measurement is questionable, but we would treat it as interval, as the 0 year would be the time when the Big Bang happened.															
17	The current BC-AD calendar was arbitrary chosen (similarly to degrees Celsius and Fahrenheit).																				
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25																					
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Categorical data (also known as qualitative data) is information that can be sorted into distinct groups or labels rather than measured numerically. It describes characteristics (like colors, brands, or survey responses) rather than quantities.

Nominal Data (a type of categorical data): Categories without a specific numerical order or hierarchy.Examples: Eye color (blue, green, brown), gender, car makes.

Numerical data, also known as quantitative data, is information expressed in numbers rather than natural language, used for measurement and arithmetic operations. Common examples include heights, weights, prices, and counts, which are crucial for analysis. It is

Discrete data is quantitative information that can only take on specific, distinct, and countable values rather than a range. You typically obtain it by counting things, and it cannot be subdivided into infinite fractions.

Interval data is quantitative information measured on a scale where the distance between each value is equal, but there is no absolute (true) zero. Because it lacks a meaningful zero, you can add and subtract the values, but you cannot multiply or divide them (e.g.,

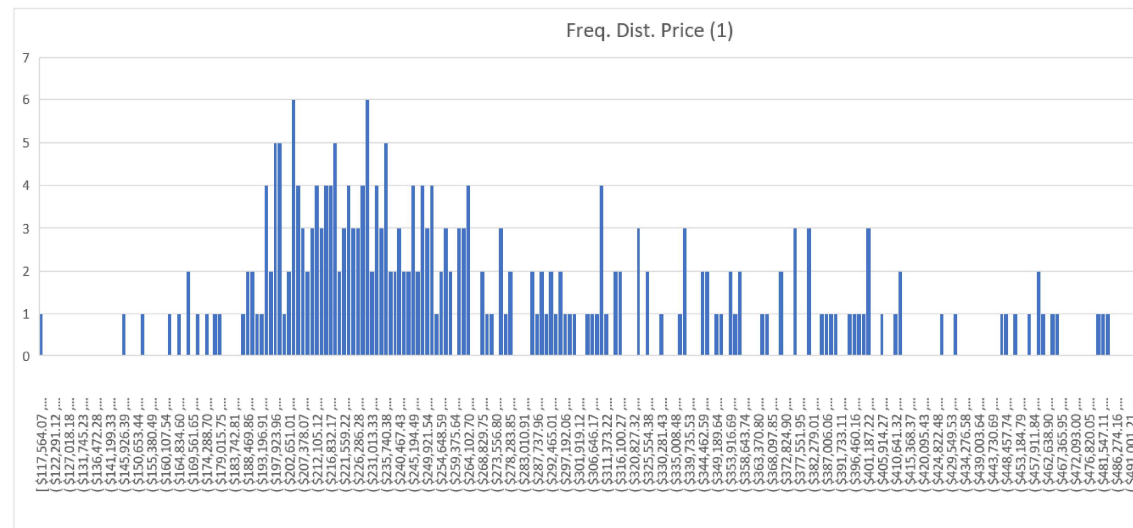
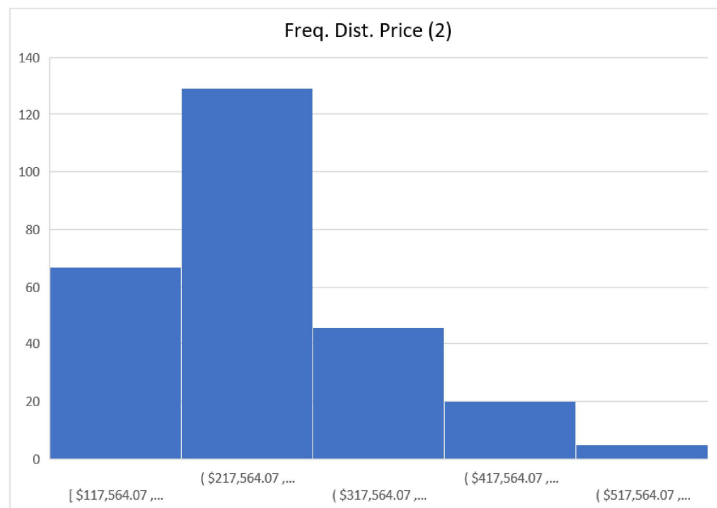
Histograms, Graphing numerical data

Task 2: Create a frequency distribution graph representing the price (that is a histogram with the highest possible number of bins - 267). Use data on all apartments and offices, no matter if sold or not.

Task 3: Create a histogram which represents the Price variable. Choose interval width (bins) of length \$100,000. If you don't know how to do that, refer to the Course notes on descriptive statistics provided with the first lecture in this section.

Use the data on all properties, no matter if sold or not.

Task 4: Interpret the results.



Graph (2): Property prices are right skewed (meaning a few very expensive properties but the majority at around the \$217K mark) Graph (1): Most of the property prices are concentrated around the \$199k to \$292K mark

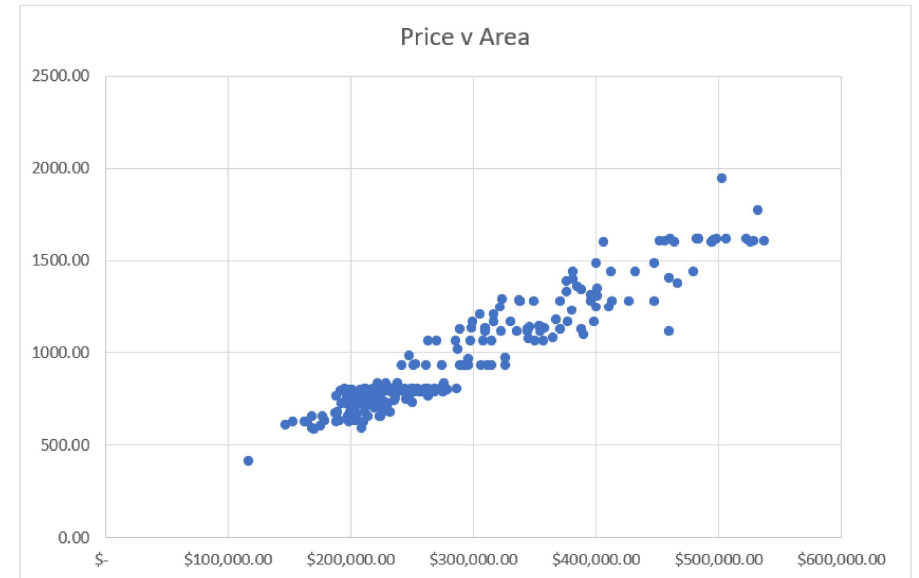
11

365 DataScience RE California Database

Scatter plot. Plotting numerical data; establishing relationship between variables.

Task 5: Create a scatter plot showing the relationship between Price and Area. Use the data on all properties, no matter if sold or not. Interpret the results.

Output (price) should always be on they-axis



There is a high correlation between the area and price

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Frequency distribution table. Pareto diagram.

Task 6: Create a frequency distribution table, where you list all the countries from which the company has buyers (country vs number of buyers). Count the absolute frequency, the relative frequency and the cumulative frequency.

Task 7: Create a Pareto diagram representing the data.

Country	Absolute Frequency	Relative Frequency	Cumulative Frequency
USA	177	91%	91%
Canada	7	4%	94%
Russia	4	2%	96%
Belgium	2	1%	97%
UK	2	1%	98%
Denmark	1	1%	99%
Germany	1	1%	99%
Mexico	1	1%	100%
Sum	195	1.00	

Country by Number of Buyers

Country	Absolute Frequency	Cumulative Frequency
USA	177	91%
Canada	7	94%
Russia	4	96%
Belgium	2	97%
UK	2	98%
Denmark	1	99%
Germany	1	99%
Mexico	1	100%

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
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365 DataScience RE California Database

Measures of central tendency, asymmetry and variability

Task 8: Calculate the mean, median, mode, skewness, variance and standard deviation of Price for all properties, no matter if sold or not.

Task 9: Interpret the measures.

Mean	\$ 281,171.90
Median	\$ 249,075.66
Mode	\$ 460,001.26
Skewness	1.10
Variance	\$7,912,471,567.21
Standard deviation	\$ 88,952.07

The mean and the median are very close together indicating that the dispersion of house prices are very close together. However, this is slightly skewed by the most frequently occurring value of \$460K as demonstrated by the skewness factor.

The variance shows a wide dispersion about the mean possibly by one outlier (expensive property) or the frequently occurring price.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
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365 DataScience RE California Database

Measures of relationship between variables

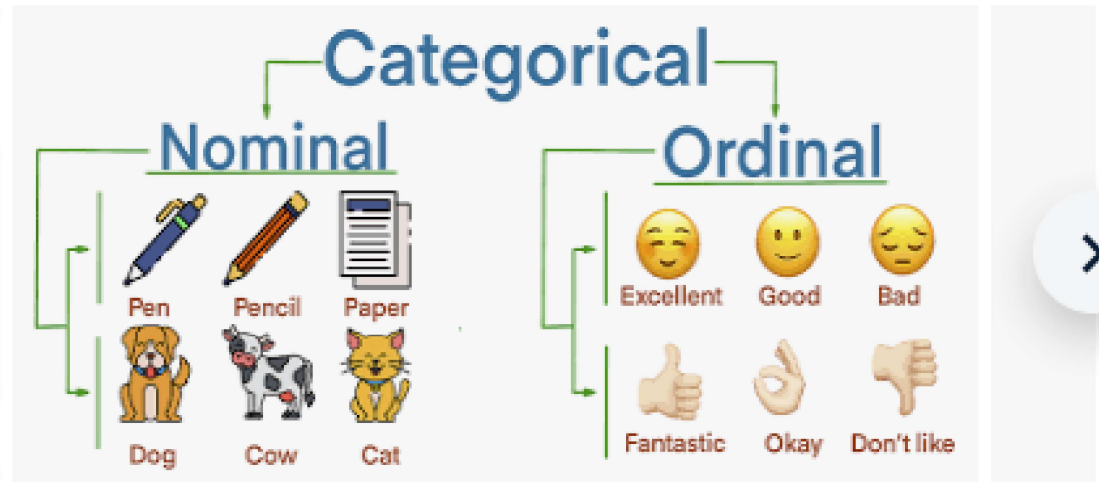
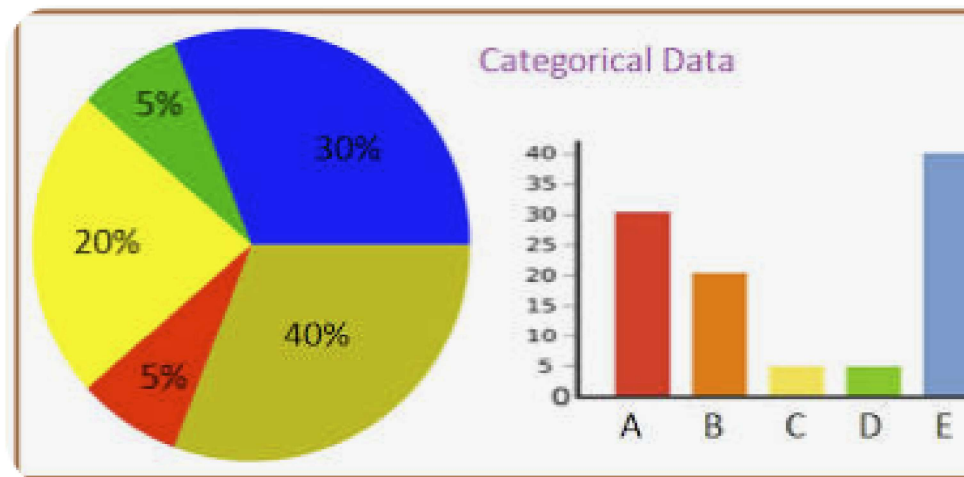
Task 10: Calculate the covariance and correlation coefficient between Price and Area, no matter if the property is sold or not. Is the result in line with the scatter plot?

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5	Covariance	\$ 24,057,280.82

7	Corrolation	0.951087377
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Correlation aligns with the scatter plot





Categorical data (also known as qualitative data) is information that can be sorted into distinct groups or labels rather than measured numerically. It describes characteristics (like colors, brands, or survey responses) rather than quantities. Fullstory +1

Main Types

- **Nominal Data:** Categories without a specific numerical order or hierarchy.
 - *Examples:* Eye color (blue, green, brown), gender, car makes.
- **Ordinal Data:** Categories that have a meaningful order or ranking, but no consistent

numerical distance between them.

- *Examples:* Customer satisfaction ratings (poor, good, excellent), education levels.  Yale University +3

How It's Analyzed

Because categorical data is non-numerical, mathematical operations like averages cannot be performed on it. Analysis primarily relies on:  Adobe +1

- **Frequencies & Counts:** Tallying how many items fall into each category.
- **Mode:** Finding the most frequently occurring category.
- **Visualizations:** Using bar charts, pie charts, or contingency tables to display groups.
- **Statistical Tests:** Utilizing tests like Chi-Square to measure associations between different groups.  Yale University +4

✦ AI Overview

Discrete data is **quantitative information that can only take on specific, distinct, and countable values rather than a range**. You typically obtain it by *counting* things, and it cannot be subdivided into infinite fractions. [Coursera +2](#)



Common Characteristics

- **Countable:** Usually answers questions like "how many?" or "how much?"
- **Fixed Intervals:** There are no possible values between distinct points (e.g., you can have 2 or 3 cars, but not 2.5 cars).
- **Distinct Categories:** It can also be qualitative (like eye color or favorite movie genres) where values fall into separate, individual groups. [Coursera +4](#)

Examples of Discrete Data

- **Whole Numbers:** The number of students in a classroom or the number of items sold in a shop.

- **Fixed Steps:** UK Shoe sizes (e.g., 5, 5.5, 6). You can be a size 5.5, but a size 5.75 does not exist.
- **Categories:** Results from a survey on preferred transport methods (e.g., Car, Bus, Train, Walk).  The Open University +4

How to Display Discrete Data

Because the values are distinct, discrete data is best represented using:

- Bar Charts
- Pie Charts
- Dot Plots
- Tally Charts  The Open University

Interval data is quantitative information measured on a scale where the distance between each value is equal, but there is no absolute (true) zero. Because it lacks a meaningful zero, you can add and subtract the values, but you cannot multiply or divide them (e.g., 10°C is not "twice as hot" as 5°C).  Dovetail +1

Key Characteristics

- **Equal Intervals:** The difference between data points is standardized and uniform.
- **No True Zero:** Zero is arbitrary and does not represent the complete absence of what is being measured.
- **Continuous Ordering:** You can rank the data logically while knowing exactly how far apart the values are.  www.scribbr.co.uk +4

Common Examples

- **Temperature:** Measured in Celsius or Fahrenheit. (Zero degrees simply means a specific point on the scale, not the "absence of temperature").

- **Calendar Dates and Times:** The time difference between 2:00 PM and 3:00 PM is the exact same as 5:00 PM and 6:00 PM.
- **Standardized Testing Scores:** IQ scores or SAT scores (a score of 0 does not mean the absence of intelligence or knowledge). www.scribbr.co.uk +4

How You Can Use It

Interval data is highly flexible for statistical analysis. You can easily calculate: [Dovetail](#)

- Central tendency: [Mean, Median, and Mode](#)
- Dispersion: Standard deviation and variance [Dovetail](#) +2

For a deeper dive into the exact statistical differences between interval and ratio scales, explore this [Scribbr Guide on Interval Data](#).